# Certified $H_\infty$ co-design of actuator, sensor and controller for the linearised Ginzburg–Landau equation

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**Abstract**

We formulate and solve a simultaneous ("co-design") optimisation of the actuator distribution, the sensor distribution and a fixed-order dynamic compensator, so as to minimise the closed-loop $H_\infty$ norm of the energy resolvent of the linearised complex Ginzburg–Landau (GL) equation in its subcritical, globally stable regime. The study is a methodological stepping stone towards flow control of a two-dimensional open cavity at $Re = 7500$: every algorithmic ingredient is chosen so that it survives the transition to a large-scale discretisation, and the GL benchmark is used only to validate it against reference solutions that remain computable by dense linear algebra.

Three results stand out. First, the disturbance amplification of the uncontrolled system, $\gamma_{\mathrm{ol}} \simeq 881.6$, is reduced by a factor of $\simeq 100$ by optimising the actuator and sensor *positions* together with the compensator, and by a factor of $\simeq 240$ when the actuator and sensor *spatial distributions* are optimised as free fields. Second, shaping the support of the actuation and of the measurement is by a wide margin the dominant effect: it is worth roughly twice as much as position optimisation, and roughly an order of magnitude more than compensator retuning alone. Third — and this is a methodological result of general relevance — a locally convergent evaluation of the $H_\infty$ norm inside the optimisation loop is *not* safe: it silently tracks a local maximiser of $\sigma_1(\omega)$ and the optimiser then minimises the wrong functional. Certification in the sense of Schwerdtner, Mengi & Voigt was necessary to obtain the results reported here.

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# 1 Motivation

## 1.1 Why co-design

Classical flow-control practice separates two decisions that are in fact coupled. The actuator and the sensor are first placed, usually guided by physical reasoning or by an open-loop sensitivity or resolvent analysis; a compensator is then synthesised for the resulting plant. The separation is convenient but arbitrary: the achievable closed-loop performance is a property of the *triple* (actuator, sensor, compensator), and there is no reason for the placement that is optimal in an open-loop sense to be optimal once feedback is closed.

Chen & Rowley [1] made this point quantitatively for the GL equation by optimising actuator and sensor placement for the closed-loop $H_2$ norm. The present work extends the programme in three directions:

1. the performance measure is the $H_\infty$ norm — the worst-case energy amplification over frequency — rather than the $H_2$ norm, because the objective in transition control is to suppress the *peak* response to the most amplified disturbance, not the frequency-integrated response;

2. the compensator is optimised *simultaneously* with the hardware, in a fixed-order structured form, rather than being re-synthesised optimally for each candidate placement;
3. the actuator and sensor are allowed to be arbitrary spatial *fields* of unit $L^2$ norm, not merely translated Gaussians of prescribed width.

### 1.2 Why the Ginzburg–Landau equation

The linearised complex GL equation reproduces the essential difficulty of the target application — strong non-normal, convective amplification of disturbances in an open shear flow — at a cost that permits exhaustive validation. Every quantity computed by the scalable algorithm can be checked against a dense reference on the same discretisation. This is what makes GL the right place to debug a methodology whose failure modes, as will be seen, are subtle and silent.

### 1.3 Requirements inherited from the cavity

The algorithm is constrained *a priori* by what will be affordable at $Re = 7500$ in two dimensions:

- **no dense linear algebra** in the optimisation loop: no explicit inverse, no dense eigen- or singular-value decomposition, no factorisation of the mass matrix;
- **the mass matrix appears only through matrix–vector products** — in particular neither $M^{-1}$ nor $M^{\pm 1/2}$ may ever be formed, since a Cholesky factor of $M$ destroys sparsity;
- **one sparse LU factorisation per frequency**, reused for the direct and the adjoint problem;
- **all eigenvalue and singular-value problems solved by Krylov methods** (Arnoldi / Krylov–Schur) on matrix-free operators;
- **certified global optimality** of the computed $H_\infty$ norm, because a local evaluation is unreliable (§3.6).

These requirements are not cosmetic. Each of them changed the algorithm in a way that is documented below.

## 2 Problem formulation

### 2.1 Governing equations and discretisation

The linearised complex Ginzburg–Landau equation is written in weak form on a P2 finite-element discretisation of the interval, giving

$$M\dot{q} = Aq + b\,u + Mf, \qquad y = c^H q, \tag{1}$$

where $M$ is the (real, symmetric, positive-definite) mass matrix, $A$ the complex GL operator, $b$ the actuator load vector, $c$ the sensor functional and $f$ a distributed disturbance. The parameters are those of Chen & Rowley: $U = 2$, $c_u = 0.2$, $c_d = -1$, $\mu_2 = -0.01$, $\mu_0 = 0.38$, which places the system in the globally *stable*, convectively amplifying regime (leading eigenvalue $-0.0177 - 0.648\,\mathrm{i}$). The nominal actuator and sensor are Gaussians of width $\sigma = 0.4$ centred at $x_a = -1.03$ and $x_s = 0.98$, the $H_2$-optimal placement of Chen & Rowley. Production runs use $n_e = 200$ elements ($n = 399$ degrees of freedom); mesh convergence is documented in §4.1.

## 2.2 The energy resolvent, and a correction to the earlier formulation

The exogenous input and the performance output are defined as follows. The disturbance is the distributed body force $f$ and the performance output is the state $q$, both measured in the energy norm $\|\cdot\|_M$. To make the induced gain an ordinary $L^2 \to L^2$ operator norm we introduce the *normalised* exogenous variables

$$w = M^{1/2} f, \qquad z = M^{1/2} q, \qquad \text{so that} \qquad \|w\|_2 = \|f\|_{L^2}, \quad \|z\|_2 = \|q\|_{L^2}, \tag{2}$$

i.e. $w$ replaces $f$, and $z$ is the energy-weighted state. Substituting $f = M^{-1/2} w$ into $M\dot{q} = Aq + Mf$, and $z = M^{1/2} q$, gives the input and output operators

$$B_w = M^{1/2}, \qquad C_z = M^{1/2} \tag{3}$$

— the mass matrix appearing on *both* sides — and hence the transfer function and its worst-case gain

$$G(s) = C_z (sM - A)^{-1} B_w = M^{1/2} (sM - A)^{-1} M^{1/2}, \qquad \gamma = \sup_{\omega} \sigma_1(G(\mathrm{i}\omega)) \,. \tag{4}$$

**The mass matrix appears on both sides.** This is a correction with respect to the formulation used in the preliminary internship report, which took $B_w = M^{-1/2}$ and $C_z = M^{1/2}$. That choice is the correct one for a *collocation* discretisation, where the disturbance is prescribed pointwise; it is not correct for a finite-element weak form, in which the disturbance enters as the *load vector* $Mf$. The consequence is not a constant offset but a loss of mesh convergence: with $B_w = M^{-1/2}$ the computed norm grows like $1/h$, doubling with each mesh refinement (1772, 3510, 7030, 14069 for $n_e = 100, 200, 400, 800$), whereas the mass-consistent form converges to $\gamma_{\mathrm{ol}} = 881.6$ at $\omega^* = -0.6458$. The value was confirmed independently on a Hermite spectral discretisation to 0.09 %. The qualitative conclusions of the earlier report, and the peak frequency $\omega^*$, are unaffected; the absolute numbers quoted there should be regarded as mesh-dependent artefacts.

**Formulation without $M^{\pm 1/2}$.** Writing $S(\omega) = \mathrm{i}\omega M - A$ and substituting the load vector $v = Mf$ as the working variable, the maximum singular value satisfies

$$\sigma^2 = \max_{f} \frac{q^H M q}{f^H M f} \quad \Longleftrightarrow \quad W v = \lambda v, \qquad W = M\, S^{-H} M\, S^{-1}, \qquad \sigma = \sqrt{\lambda}. \tag{5}$$

This is a *standard* eigenvalue problem. Applying $W$ costs one solve with $S$, one multiplication by $M$, one solve with $S^H$ and one further multiplication by $M$. The operator $W$ is self-adjoint in the $M^{-1}$ inner product, so $\lambda$ is real and positive, but $M^{-1}$ never appears in the algorithm. The only factorisation is the sparse LU of $S$, which preserves the bandwidth of the discretisation (4.0 non-zeros per row at every mesh) and is reused for both $S$ and $S^H$.

## 2.3 The co-design problem

With the compensator of order $r$ in state-space form $(A_c, B_c, C_c, D_c)$, the closed loop is the descriptor system

$$M_{cl} \dot{\xi} = A_{cl}(\theta) \xi + \begin{bmatrix} Mf \\ 0 \end{bmatrix}, \qquad M_{cl} = \mathrm{diag}(M, I_r), \tag{6}$$

and the design problem is

$$\min_{\theta} \ \gamma(\theta) = \max_{\omega} \ \sigma_1\big(C_z (\mathrm{i}\omega M_{cl} - A_{cl})^{-1} B_w\big) \qquad \text{s.t.} \qquad \alpha(A_{cl}, M_{cl}) + \varepsilon \le 0, \tag{7}$$

where $\alpha$ is the spectral abscissa of the closed-loop pencil and $\varepsilon = 10^{-3}$ enforces a margin of internal stability. The design vector $\theta$ collects any subset of the groups in Table 1.

| token | design variables | count |
|---|---|---|
| `K` | compensator $(d, e, f, n, D_c)$ | $4r - 1$ complex |
| `B` | actuator centre $x_a$ | 1 real |
| `C` | sensor centre $x_s$ | 1 real |
| `FB` | actuator field $\psi_a$, $\|\psi_a\|_{L^2} = 1$ | $n$ complex |
| `FC` | sensor field $\psi_s$, $\|\psi_s\|_{L^2} = 1$ | $n$ complex |

**Table 1:** Design-variable groups. A case is denoted by the "+"-separated list of the groups it frees, e.g. `K+FB+FC`.

The nested structure of these cases provides a strong correctness check: each case's feasible set strictly contains those of its "parents", so its optimal $\gamma$ can never be larger. This monotonicity is verified automatically and is reported in §4.

# 3 Methodology

The guiding philosophy is that *every quantity used by the optimiser must be computable at cavity scale, and every quantity reported must be certified.* The two are independent requirements and both proved necessary.

## 3.1 Closed loop as a low-rank correction

The compensator couples to the plant through a single actuator and a single sensor, so the closed-loop operator is a **rank-two correction** of a block-diagonal one:

$$S_{cl}(s) = \underbrace{\begin{bmatrix} sM - A & 0 \\ 0 & sI_r - A_c \end{bmatrix}}_{S_0} + UV, \qquad U = \begin{bmatrix} b & 0 \\ 0 & B_c \end{bmatrix}, \qquad V = -\begin{bmatrix} D_c\, c^H & C_c \\ c^H & 0 \end{bmatrix}, \tag{8}$$

with $U \in \mathbb{C}^{(n+r)\times 2}$, $V \in \mathbb{C}^{2\times(n+r)}$: the first column of $U$ and the first row of $V$ carry the actuator $b$ and the control law, the second pair the sensor $c$ and the compensator input $B_c$. The closed-loop transfer function from $w$ to $z$ is

$$G_{cl}(s) = \mathcal{C}_z\, S_{cl}(s)^{-1} \mathcal{B}_w, \qquad \mathcal{B}_w = \begin{bmatrix} M^{1/2} \\ 0 \end{bmatrix}, \qquad \mathcal{C}_z = \begin{bmatrix} M^{1/2} & 0 \end{bmatrix}, \tag{9}$$

the disturbance entering, and the performance output being read from, the *plant* block only. This observation is what fixes the certification pencil of §3.6: $\mathcal{B}_w \mathcal{B}_w^H = \mathcal{C}_z^H \mathcal{C}_z = \mathrm{diag}(M, 0)$, not $M_{cl}$. The Sherman–Morrison–Woodbury identity then gives every closed-loop solve in terms of one sparse LU of $sM - A$ and a $2 \times 2$ capacitance matrix,

$$S_{cl}^{-1} = S_0^{-1} - Y G^{-1} V S_0^{-1}, \qquad Y = S_0^{-1} U, \qquad G = I_2 + VY, \tag{10}$$

with the adjoint obtained from the same factors. The augmented (state-space) form is retained deliberately in preference to the algebraically simpler rank-one condensation $\left[sM - A - \kappa(s) b c^H\right] q = Mf$: the augmented pencil is *linear* in $s$, which is required for the stability eigenproblem and for the certification pencil of §3.6; it accommodates a singular $E$ (the pressure block of the cavity); and it extends to MIMO by widening $U$ and $V$ to $(n + r) \times (m + p)$ without any change of algebra.

The spectral abscissa is obtained by shift-and-invert Krylov iteration on the same machinery, since $A_{cl} - \sigma M_{cl} = -S_{cl}(\sigma)$; left eigenvectors come from the adjoint operator using the same factors. This replaced a dense generalised eigensolve and reduced its cost from 1.93 s to 0.05 s at $n_e = 200$, making the whole oracle essentially mesh-independent ($\simeq 0.25$ s per call from $n = 199$ to $n = 1599$).

### 3.2 Compensator parametrisation: tridiagonal, not canonical

The compensator was initially parametrised in complex controllable canonical (companion) form, which is minimal: $2r + 1$ complex parameters. This proved to be a serious numerical error, and the diagnosis is worth recording.

The *spectrum* of a realisation is invariant, but its *conditioning* is not. The eigenvector matrix of a companion matrix is a Vandermonde matrix in the poles: if $p_1, \ldots, p_r$ are the roots of its characteristic polynomial, the eigenvector for $p_j$ is $(1, p_j, p_j^2, \ldots, p_j^{r-1})^T$, so the eigenvector matrix is $V_{ij} = p_j^{i-1}$ — the Vandermonde matrix built on the poles. Its condition number grows exponentially when the nodes span several decades. For the compensators arising here the poles span $|p| \in [0.49, 3883]$, a ratio of $7.9 \times 10^3$, and the resulting eigenvector condition number is $6.5 \times 10^4$. Worse, the map from poles to polynomial coefficients is Wilkinson-ill-conditioned, so the companion coefficients reach $4.4 \times 10^8$. The consequences were concrete: the certification pencil of §3.6 returned meaningless eigenvalues, and the optimiser's steps were badly scaled.

The remedy is a **tridiagonal** parametrisation $A_c = \mathrm{tridiag}(f, d, e)$ with the input gauge $B_c = \mathbf{1}$, using $4r - 1$ complex parameters. Measured on the same compensator, the eigenvector condition number falls to $\mathcal{O}(1)$ and the largest coefficient to $1.2 \times 10^5$. Three properties make this the right choice:

- it **contains the modal (diagonal) case** $e = f = 0$, which is how seeds are initialised — exactly, by an eigendecomposition of the small *balanced* reduced realisation, so that a companion matrix is never formed at any stage;
- unlike a real modal form, it represents real poles and complex-conjugate pairs with a **single smooth parametrisation**, so eigenvalues can migrate on and off the real axis continuously — a property that will be needed for the real cavity operator;
- it generalises to MIMO by widening $B_c$ to $r \times m$ and $C_c$ to $p \times r$.

The $\simeq 2r$ redundant parameters are negligible beside the $\mathcal{O}(n)$ field unknowns, and buy a conditioning improvement of four orders of magnitude. Conversion was validated to $1.1 \times 10^{-14}$ on the transfer function and exactly on the spectrum.

### 3.3 Field parametrisation and Sobolev regularisation

The actuator and sensor fields are parametrised through the retraction $\psi = v/\|v\|_M$, so that $\|\psi\|_{L^2} = 1$ holds *exactly* at every iterate rather than asymptotically. Since $b \to b/\alpha$ can be absorbed by the compensator, the constraint fixes a gauge rather than limiting control authority; the compensator is rescaled accordingly so that normalisation leaves $\gamma$ invariant. The same normalisation is applied to the Gaussian placement cases, so that all designs are compared at equal $L^2$ "area under the curve" and the field plots compare *shape* only.

An unregularised field optimisation is ill-posed in a specific and instructive way. The objective sees $\psi$ only through the resolvent $(\mathrm{i}\omega M - A)^{-1}$, which strongly damps high wavenumbers: the measured sensitivity of $\gamma$ to a smooth perturbation exceeds that to a grid-scale perturbation by a factor of 50. The high-wavenumber directions are therefore a near-null-space of the objective;

the gradient there is correct but essentially zero, and the optimiser deposits grid-scale noise that costs nothing and that would be entirely different on a finer mesh.

The cure is a Sobolev ($H^1$) change of variable, $\psi = P\xi$ with $P = (M + \ell^2 K)^{-1} M$, where $K$ is the finite-element *stiffness* matrix $K_{ij} = \int \varphi_i' \varphi_j' \, \mathrm{d}x$ — the weak-form discrete Laplacian, so that $\psi^H K \psi \simeq \|\partial_x \psi\|_{L^2}^2$ — and $\ell = 0.8$ is a smoothing length. The operator $M + \ell^2 K$ therefore penalises gradients, and $\ell$ sets the scale below which structure is suppressed. A grid-scale $\xi$ maps to a smooth $\psi$, so the BFGS iteration accumulates curvature in a smooth space and the optimum is mesh-robust. Implemented as a change of variable rather than a preconditioned gradient — a one-shot preconditioning does not survive the BFGS history — this reduced the field roughness by a factor of seven while *improving* $\gamma$, confirming that the grid-scale content was never buying performance.

### 3.4 Gradients

All gradients are analytic and are obtained from the same resolvent vectors used to evaluate the objective, at no additional solve. With $x = S_{cl}^{-1} B_w v$ and $z = S_{cl}^{-H} C_z^H u$ at the active frequency,

$$\frac{\partial \sigma}{\partial \theta_k} = \mathrm{Re}\left( z^H \frac{\partial A_{cl}}{\partial \theta_k} x \right), \qquad \frac{\partial \sigma}{\partial \omega} = \mathrm{Im}\left( z^H M_{cl} x \right). \tag{11}$$

These are the standard first-order perturbation formulae for a *simple* singular value (equivalently for the corresponding eigenvalue of the operator $W$ of (5)); the frequency derivative is Eq. (15) of Benner & Mitchell [4], and the design derivative follows from differentiating $\sigma = u^H G v$ with $u, v$ held fixed to first order. They require the maximal singular value to be simple, which is monitored: the singular-value gap is returned by every oracle call. Each structural derivative $\partial A_{cl}/\partial \theta_k$ is a single sparse outer product, so each gradient component is a scalar product. Every gradient family was validated against finite differences: compensator $3.4 \times 10^{-7}$, actuator and sensor placement $\sim 10^{-8}$, actuator and sensor fields $5 \times 10^{-10}$. The spectral-abscissa gradient agrees with an eigenvalue-*tracked* finite difference to $10^{-6}$; the residual disagreement observed at some iterates against a naive finite difference is a genuine rightmost-eigenvalue crossing, at which $\alpha$ is non-smooth and the analytic gradient is a one-sided Clarke subgradient.

### 3.5 Optimisation

The non-smooth, non-convex constrained problem (7) is solved with pyGRANSO [7] (BFGS-SQP with gradient sampling, Lewis–Overton line search), which is designed for exactly this class: $\gamma(\theta)$ is a maximum over frequency and is generically non-differentiable at the optimum, where several frequencies are active.

Two implementation points proved decisive.

**Initial Hessian calibration.** BFGS takes its first step along $-H_0^{-1} g$. With $H_0 = I$, $\|g\| \sim 10$–$80$ and a useful step of $\mathcal{O}(10^{-2})$, the first step overshoots into the unstable region and the run never recovers — the compensator cases made *zero* progress. The heuristic follows from one line of dimensional analysis: with $H_0 = hI$ the first step is $\Delta z = -g/h$, so requiring it to be a prescribed *relative* step $\|\Delta z\| = s\|z\|$ gives

$$h = \frac{\|g\|}{s\,\|z\|}, \qquad \text{i.e.} \qquad H_0 = \frac{\|g\|}{s\,\|z\|}\, I, \tag{12}$$

with $s$ the "first step" parameter. This makes the initial displacement a controlled fraction of the design vector rather than an accident of the units in which $\theta$ happens to be expressed. The

calibration is *not* performed per active frequency: $H_0$ is a single scalar multiple of the identity on the whole design vector, computed once at the start of each ladder stage from the gradient at that stage's starting point. The active frequencies enter only through the subgradients that pyGRANSO samples; they are not separate optimisation subproblems. The appropriate step size, however, depends on the local geometry: at a *smooth* optimum (one active frequency) a small step is required, whereas at a *non-smooth* optimum (two or more equalised peaks) the step must be large enough to cross the kink so that BFGS accumulates curvature from both branches. This was measured: from a two-active-frequency point, a first step of 0.02 produced no progress at all while 0.3 reduced $\gamma$ from 9.3822 to 7.2123. Rather than guess, each run executes a **ladder** of first-step values $0.30 \to 0.05 \to 0.01$, each stage restarting from the best feasible point of the previous one. The ladder *partitions* a fixed budget rather than multiplying it: the three stages share the wall-clock allocation of a single run, so its only cost is that BFGS curvature is discarded twice. That is affordable at cavity scale — indeed it is desirable there, because a Hessian accumulated before a change of active frequency is curvature of the wrong function.

**Diverse initialisation.** Seeding every run from the LQG-balanced-truncation compensator confines the search to the $H_2$-optimal basin, from which the only local way to reduce the $H_\infty$ *peak* is to push a closed-loop pole towards the imaginary axis, producing spuriously low $\gamma$ values with $\alpha \approx 0$ (the resolvent peak then scales as $1/|\alpha|$).

It should be said clearly that of the two remedies applied to this pathology — the tridiagonal re-parametrisation of §3.2 and the diverse seeds described here — the *re-parametrisation was the dominant one*. With the companion form the computed spectral abscissa was itself unreliable: for one and the same compensator it was returned as $\alpha = -0.101$ in companion coordinates and $\alpha = -0.225$ in tridiagonal coordinates, the latter being correct. A substantial part of the apparent boundary-hugging was therefore a *numerical artefact* of an ill-conditioned realisation, not a genuine property of the optimum. After the change of parametrisation, every case in Table 3 terminates with $\alpha$ between $-0.14$ and $-0.50$ and not one triggers the spike diagnostic. The diverse seeds are retained because the basin argument remains valid in principle and they cost little, but on this benchmark they were not what removed the problem. Seven structurally distinct compensators are therefore constructed, all balanced-truncated to a common order $r = 7$ and converted to tridiagonal form: the full-field LQG-BT compensator; three LQG compensators; and three $H_\infty$ central (DGKF) compensators. The latter six are synthesised with **low-rank exogenous channels**, $w = \phi\, a(t)$ and $z = \psi^T x$ with $\phi, \psi$ Gaussians placed at $(-1, 1)$, $(-2, 2)$, $(-3, 3)$ — cheap enough to be carried to the cavity, unlike a full-field $H_\infty$ Riccati equation. The objective itself is unchanged; the seeds only select the starting basin.

### 3.6 Certification: the essential ingredient

The frequency search inside the optimisation loop is a multistart *local* method: a coarse sweep, seeds derived from the least-damped closed-loop eigenvalues, and warm-started active frequencies, each refined by a *safeguarded* Newton step — Newton on $\partial\sigma/\partial\omega = 0$, protected by a trust region on $|\Delta\omega|$, by rejection of the step when the finite-difference curvature is not negative (so that it cannot converge to a *minimum* of $\sigma_1$), and by backtracking until $\sigma_1$ actually increases. It is fast and adequate as an optimisation oracle, but it carries **no guarantee** of finding the global maximum of $\sigma_1(\omega)$.

This is not an academic concern. It was observed directly in this study: an optimiser reporting $\gamma = 1.6675$ for a design whose true norm was 10.4574; and, more insidiously, field cases in which the reported $\gamma$ *decreased* from one run to the next while the true norm *increased* — the optimiser had flattened the peak it was tracking and grown a narrow peak elsewhere, and was then minimising a quantity that was no longer the $H_\infty$ norm. This is precisely the failure

illustrated in Fig. 1 of Schwerdtner, Mengi & Voigt [5]. A denser frequency sweep does not cure it: on random field perturbations a nine-point sweep is already accurate to 0.01 %; the problem appears only at *optimised* configurations, where the optimiser has reshaped the gain curve.

We therefore implemented the certification algorithm of [5] in a fully matrix-free form. It is a *decision procedure*: given $\gamma$, it determines whether $\|H\|_\infty > \gamma$. Its basis is that $\gamma$ is a singular value of the resolvent at frequency $\omega$ if and only if $\lambda = \mathrm{i}\omega$ is an eigenvalue of the even pencil

$$A_H = \begin{bmatrix} A_{cl} & \gamma^{-2} B_w B_w^H \\ C_z^H C_z & A_{cl}^H \end{bmatrix}, \qquad E_H = \begin{bmatrix} M_{cl} & 0 \\ 0 & -M_{cl} \end{bmatrix}. \tag{13}$$

For the energy weighting used here, $B_w B_w^H = C_z^H C_z = \mathrm{diag}(M, 0)$, which vanishes on the compensator states.

The algorithm maintains a list of critical intervals of the imaginary axis. At the midpoint of the first interval it computes the nearest pencil eigenvalues by shift-and-invert Arnoldi; if one is purely imaginary a witness has been found and $\|H\|_\infty > \gamma$; otherwise the distance to the nearest eigenvalue defines a provably **eigenvalue-free disk**, which is removed, and the two remaining subintervals are treated recursively. When no intervals remain, no purely imaginary eigenvalue exists. Only the *nearest* eigenvalue is needed per shift, and the disk radius adapts to the local spectrum.

Two aspects are specific to our implementation:

- **Sparsity is preserved.** The even pencil inherits the structure of the closed loop: it is a sparse base plus a **rank-four** correction (verified to $8.4 \times 10^{-15}$), so each shifted solve is one sparse LU of the doubled base pencil plus a $4 \times 4$ capacitance, with the same factors used for the direct and adjoint problems.
- **The full band must be scanned.** The original algorithm starts at $\omega = 0$, exploiting the spectral symmetry of a *real* system. The GL operator is complex, that symmetry does not hold, and the interval $[\omega_{\mathrm{lo}}, \omega_{\mathrm{hi}}]$ must be covered in full. The band is determined automatically from an open-loop sweep plus a margin. This restriction will *not* apply to the (real) cavity operator.

We did **not** implement the structure-preserving spectral transformation of Mehrmann & Watkins [6] used in the reference. Its purpose is to guarantee that imaginary eigenvalues emerge with exactly zero real part, removing the need for a tolerance. In our pencil the measured separation is $|\operatorname{Re}\lambda| \sim 10^{-17}$–$10^{-14}$ for on-axis eigenvalues against $1.2 \times 10^{-1}$ for the nearest off-axis ones — thirteen orders of magnitude, holding even when $\gamma$ is within $10^{-4}$ of the peak and two crossings nearly merge. A plain relative tolerance is unambiguous, and the transformation's additional cost is not justified here. (This separation is a *consequence* of the tridiagonal parametrisation of §3.2; with companion coefficients of $10^8$ the test was not reliable.)

Certification is applied in two places: at the end of every ladder stage, where a discovered witness frequency is injected into the warm-start seeds so that the next stage tracks the correct peak; and as a post-processing pass over all results. From a good lower bound — which the local search provides for free — the certificate converges in two calls, taking 0.1–5 s per design. The cheap local method and the certified global method are thus genuinely complementary: the former supplies the tight lower bound that makes the latter cheap, and the latter repairs the former when it is tracking the wrong peak.

# 4 Results

## 4.1 Verification

The checks in Table 2 were carried out before any design was computed. Mesh convergence of the closed-loop norm is four significant figures across a fourfold refinement (10.7959, 10.7983, 10.8021 for $n_e = 100, 150, 200$), so $n_e = 200$ is used throughout.

| quantity | check | agreement |
|---|---|---|
| sparse closed-loop gain | vs dense reference | $4 \times 10^{-14}$ |
| Krylov spectral abscissa | vs dense generalised eigensolve | $9 \times 10^{-16}$ |
| tridiagonal conversion | transfer function; spectrum | $1.1 \times 10^{-14}$; exact |
| compensator gradients | vs finite differences | $3.4 \times 10^{-7}$ |
| placement gradients | vs finite differences | $\sim 10^{-8}$ |
| field gradients | vs finite differences | $5 \times 10^{-10}$ |
| certified norm | vs 4001-point sweep | $5 \times 10^{-6}$ |
| decision procedure | protocol of [5] | 6/6 correct |

**Table 2:** Verification of the matrix-free implementation against dense references and finite differences.

## 4.2 Optimal designs

All values in Table 3 are **certified**: every case was verified by the decision procedure of §3.6 to be the global maximum of $\sigma_1(\omega)$ over the band, to a relative tolerance of $10^{-4}$. All nesting relations are satisfied, no design converged onto the stability boundary, and every case is converged.

| group | design | $\gamma_\infty$ | $\alpha$ | act. freq. | reduction |
|---|---|---|---|---|---|
| – | uncontrolled | 858.20 | −0.018 | 1 | — |
| | LQG-BT (reference) | 10.80 | — | 1 | ×79 |
| placement | optim `B` | 10.1375 | −0.193 | 2 | ×85 |
| | optim `C` | 10.0109 | −0.195 | 2 | ×86 |
| | optim `B+C` | 9.6757 | −0.202 | 2 | ×89 |
| compensator structure | `KG` static real gain | 9.8896 | −0.219 | 1 | ×87 |
| | `KGC` static complex gain | **9.8319** | −0.211 | 1 | ×87 |
| | `KSP` strictly proper | **9.8319** | −0.226 | 1 | ×87 |
| | `K` proper, $r = 7$ | **9.8319** | −0.226 | 1 | ×87 |
| compensator + placement | optim `K+B` | **9.8319** | −0.226 | 1 | ×87 |
| | optim `K+C` | **9.6601** | −0.229 | 1 | ×89 |
| | optim `K+B+C` | 8.4643 | −0.251 | 1 | ×101 |
| REAL fields | optim `FBR` | **9.8319** | −0.153 | 1 | ×87 |
| | optim `FCR` | **9.6601** | −0.211 | 2 | ×89 |
| | optim `FBR+FCR` | 5.6097 | −0.215 | 2 | ×153 |
| | optim `KG+FBR+FCR` | 4.1971 | −0.211 | 1 | ×204 |
| | optim `K+FBR+FCR` | *4.1140* | −0.132 | 1 | ×209 |
| COMPLEX fields | optim `FB` | **9.8319** | −0.208 | 1 | ×87 |
| | optim `FC` | **9.6601** | −0.142 | 1 | ×89 |
| | optim `FB+FC` | 4.3807 | −0.217 | 4 | ×196 |
| | optim `K+FB+FC` | **3.6741** | −0.499 | 1 | ×**234** |

**Table 3:** Certified optimal designs at $n_e = 200$, compensator order $r = 7$. Values in **bold** sit on a fundamental bound of §4.3: 9.8319 is the sensor-limited bound for $x_s = 0.98$ and 9.6601 the actuator-limited bound for $x_a = -1.03$. *Every* case is certified to be the global maximum of $\sigma_1(\omega)$ over the band, every nesting relation is satisfied, and no design terminates on the stability boundary.

The dominant conclusion is the ordering of the three design freedoms. Relative to the LQG-BT reference:

- retuning the **compensator** alone buys 9 %;
- optimising actuator and sensor **positions** buys 10 %, and together with the compensator 22 %;
- optimising the actuator and sensor **spatial distributions** buys 59 %, and with the compensator 66 %.

Shaping the support of the actuation and of the measurement is thus worth roughly twice as much as moving it, and roughly seven times as much as retuning the compensator. The physical interpretation is that in a strongly convective system the achievable performance is limited by *how much of the amplifying region the actuator can excite and the sensor can observe*, far more than by the dynamics of the compensator connecting them.

Two secondary observations are worth recording. First, the full co-design `K+FB+FC` attains both the lowest norm and by a clear margin the deepest stability margin ($\alpha = -0.499$, twice that of any other design): the extra freedom is not spent approaching the stability boundary but on a genuinely more robust design. Second, `FB+FC` terminates with **four active frequencies**, the textbook signature of a non-smooth $H_\infty$ optimum, which is independent evidence that the field optimisation has genuinely converged rather than stalled.

### 4.3 A fundamental limit: the performance is actuator/sensor limited

Several designs were observed to coincide to six significant figures: `K`, `K+B`, `KSP`, `KGC` and `FB` all give 9.831876, and `K+C` and `FC` both give 9.660144. Two readings were possible — the same closed loop realised in different ways (*substitution*), or different closed loops limited by the same bound (*shared active constraint*). The test of `test_substitution.py` settles it: comparing $\sigma_1(\omega)$ across the whole band gives, against `K` as reference, a relative sup-norm difference of $4.6 \times 10^{-1}$ for `FB` and $2.8 \times 10^{-1}$ for `FC` (correlations 0.94), while the peak values agree to $10^{-13}$. The closed loops are therefore *different*; only the peaks coincide.

**The explanation is a hard performance limit, and it is attained.** With one actuator and one sensor, Sherman–Morrison gives

$$G_{cl}(\mathrm{i}\omega) = T(\mathrm{i}\omega) + q(\omega)\, u(\omega)\, r(\omega)^H, \qquad u = L^H S^{-1} b, \quad r = L^H S^{-H} c, \tag{14}$$

with $T = L^H S^{-1} L$ the open-loop energy resolvent ($M = LL^H$) and $q$ a free complex scalar at each frequency — *whatever* the controller. The correction is rank one with its left factor fixed by the **actuator** and its right factor fixed by the **sensor**. A rank-one correction cannot touch the part of $T$ orthogonal to those directions, so for any controller of any order

$$\gamma \;\geq\; \sup_{\omega}\; \sigma_1(T(I - P_r)) \qquad \text{(sensor-limited: the disturbance the sensor cannot see)} \tag{15}$$

$$\gamma \;\geq\; \sup_{\omega}\; \sigma_1((I - P_u)T) \qquad \text{(actuator-limited: the response the actuator cannot reach)} \tag{16}$$

$$\gamma \;\geq\; \sup_{\omega}\; \sigma_2(T) \qquad \text{(absolute floor for } \textit{any} \text{ single actuator/sensor pair)} \tag{17}$$

where $P_r, P_u$ are the orthogonal projectors onto $r$ and $u$. Evaluated for this configuration (`bounds.py`, $n_e = 200$):

| bound | value | attained by | measured $\gamma$ |
|---|---|---|---|
| sensor-limited, $x_s = 0.98$ | 9.831865 | `K`, `K+B`, `KSP`, `KGC`, `FB` | 9.831876 |
| actuator-limited, $x_a = -1.03$ | 9.660005 | `K+C`, `FC` | 9.660144 |
| absolute floor $\sup_\omega \sigma_2$ | 3.509066 | — (best design) | 3.674055 |

**Table 4:** Fundamental limits for a single actuator and a single sensor, and the designs that attain them. The binding bound is the larger of the first two.

The agreement is to $1.1 \times 10^{-5}$ and $1.4 \times 10^{-4}$ respectively, and the argmax frequencies coincide with the optima ($\omega^* = -0.53$). The whole pattern of Table 3 follows:

- with the sensor held at $x_s = 0.98$ the sensor bound 9.8319 exceeds the actuator bound and therefore binds. *Nothing* improves on it — not a seventh-order compensator, not moving the actuator, not even optimising the actuator as a 798-parameter field. All five designs sit on the bound;
- once the sensor is freed, the optimiser lowers the sensor bound until the *actuator* bound 9.6600 takes over, which is why `K+C` and `FC` — two quite different closed loops — again share a value;
- only when both are free do both bounds fall, and the best design reaches 3.674, within 4.7 % of the absolute floor 3.509 that no single actuator/sensor pair can beat.

**This is the central result of the study. For this configuration the achievable $H_\infty$ performance is set almost entirely by *what the actuator can reach and what the***

***sensor can see*; the compensator is very nearly irrelevant, and the co-design gains reported in Table 3 are gains in observability and controllability, not in control law.** The bounds are computed *matrix-free* (`bounds.py`): one sparse LU of $S = \mathrm{i}\omega M - A$ per frequency, reused for the direct and adjoint solves, with $M$ only as a mat-vec, and each bound obtained as the largest eigenvalue of $W = MR^H MR$ for a rank-one-deflated solve operator $R$. A dense reference (`-method dense`, forming $T$ explicitly) agrees to $2 \times 10^{-4}$ at every frequency but scales as $\mathcal{O}(n^3)$:

| | $n_e = 200$ | $n_e = 800$ |
|---|---|---|
| dense | 6.8 s | 180.6 s |
| matrix-free | **0.7 s** | **1.2 s** |

a factor 153 already at $n = 1599$, and the gap widens as $n^2$. The bounds are independent of any optimisation and should be computed *first* in future studies: they say in advance how much a given hardware layout can possibly deliver, and they certify how close a converged design is to the best possible.

### 4.4 Controller structure and the realness of the fields

Two aspects of the comparison in Table 3 deserve scrutiny, and both were raised in review.

**The field comparison was not like-for-like.** The nominal actuator and sensor are *real* Gaussians, whereas the optimised fields of §3.3 were allowed to be complex. A complex field carries twice as many degrees of freedom, so part of the improvement attributed to "shaping" is in fact attributable to "allowing a complex profile" — a different, and physically much more demanding, statement about what the hardware must do. The comparison has therefore been split: the cases `FBR`, `FCR` and `FBR+FCR` constrain the field to be real (only $\mathrm{Re}\,\psi$ is a design variable, $\mathrm{Im}\,\psi \equiv 0$), which is the fair comparison against a real Gaussian; the existing complex cases then measure the additional value of a complex profile on top of that. Figure 2 has accordingly been redrawn as a $2 \times 2$ array showing $\mathrm{Re}\,\psi$ and $\mathrm{Im}\,\psi$ separately, so that the extra degree of freedom is visible rather than hidden inside a modulus.

**How much dynamics does the compensator actually need?** $H_\infty$-optimal compensators for this kind of problem are often close to static gains, and it is worth measuring how much the seventh-order dynamics buy. Four nested structures are now available, each freeing strictly more parameters than the last:

| token | structure | free parameters |
|---|---|---|
| `KG` | static *real* gain, $u = D_c y$, $D_c \in \mathbb{R}$ | 1 real |
| `KGC` | static *complex* gain, $D_c \in \mathbb{C}$ | 2 real |
| `KSP` | dynamic, strictly proper ($D_c \equiv 0$) | $2(4r-2)$ real |
| `K` | dynamic, proper (dynamics *and* $D_c$) | $2(4r-1)$ real |

We note in passing that the compensator used throughout this study was *already* proper: $D_c$ is a design variable, initialised at zero and free to move. The `KSP` case, which pins $D_c \equiv 0$, therefore isolates the contribution of the feedthrough term rather than adding one.

The production results ($n_e = 200$, certified) are unambiguous:

| structure | free parameters | $\gamma_\infty$ | |
|---|---|---|---|
| `KG` (static *real* gain) | 1 | 9.8896 | 0.6 % above the bound |
| `KGC` (static *complex* gain) | 2 | 9.8319 | **on the bound** |
| `KSP` (strictly proper, $r = 7$) | 52 | 9.8319 | on the bound |
| `K` (proper, $r = 7$) | 54 | 9.8319 | on the bound |

**Two complex numbers — a single static complex gain — attain exactly the same $\gamma$ as a fifty-four-parameter seventh-order compensator, to thirteen significant figures.** Adding dynamics buys nothing whatsoever. Restricting the gain to be *real* does cost something (9.8896 versus 9.8319, i.e. 0.6 %), and §4.3 explains why: the optimal rank-one correction requires a complex coefficient at each frequency, and a real gain cannot supply one.

This is corroborated independently by the Bode diagrams of §4.7: the *dynamic* strictly-proper compensator `KSP`, with fifty-two free parameters, converges to a transfer function that varies by 0.73 dB over the entire band $\omega \in [-4, 4]$ — it has become a constant of its own accord.

## 4.5 Frequency response

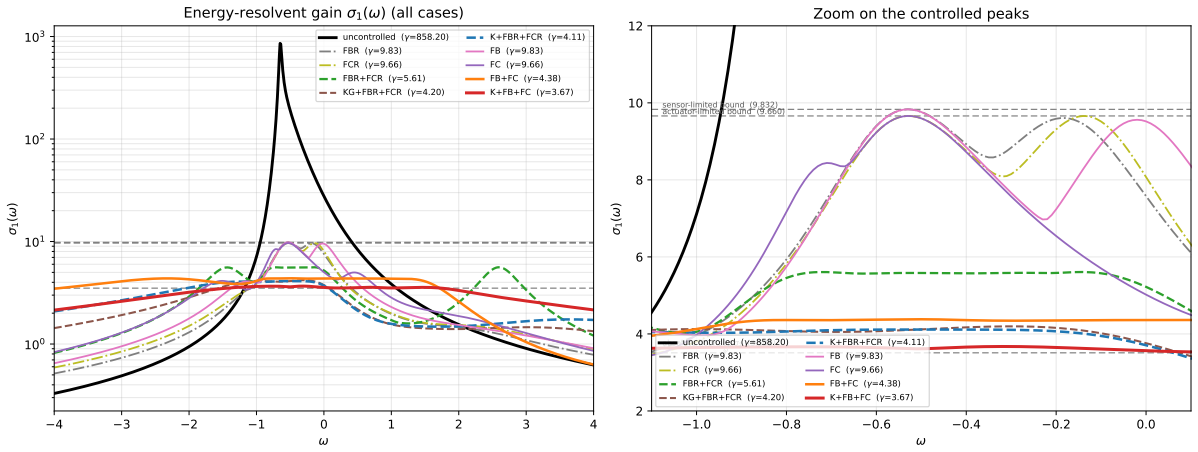



**Figure 1:** Energy-resolvent gain $\sigma_1(\omega)$ for all designs. Left: logarithmic scale over the full band. Right: zoom on the controlled peaks. The uncontrolled resonance near $\omega^* = -0.65$ is three orders of magnitude above every controlled design. The field designs (`FB+FC`, `K+FB+FC`) are qualitatively flatter across the band, not merely lower.

The uncontrolled response (Fig. 1) is dominated by a single very sharp resonance near $\omega^* = -0.65$. Its sharpness is itself instructive: the value labelled in the figure (858) is 3 % below the true norm (881.6) simply because an 801-point sweep does not resolve the peak — a direct illustration, on the simplest possible case, of why the certification of §3.6 is necessary.

All controlled designs collapse this resonance by two orders of magnitude and replace it by a much broader, flatter response. Among them the ordering of Table 3 is clearly visible: the placement and compensator designs form a tight cluster around $\gamma \simeq 9.7$–$10.1$ with a residual peak still centred near $\omega \simeq -0.53$, essentially the plant resonance attenuated; `K+B+C` detaches from that cluster at 8.46; and the two field designs are qualitatively different, with a markedly *flatter* response over the whole band and peaks at 4.38 and 3.67. The field designs do not merely attenuate the plant resonance — they redistribute the gain, which is why they reach a substantially lower worst case.

## 4.6 Optimal distributions

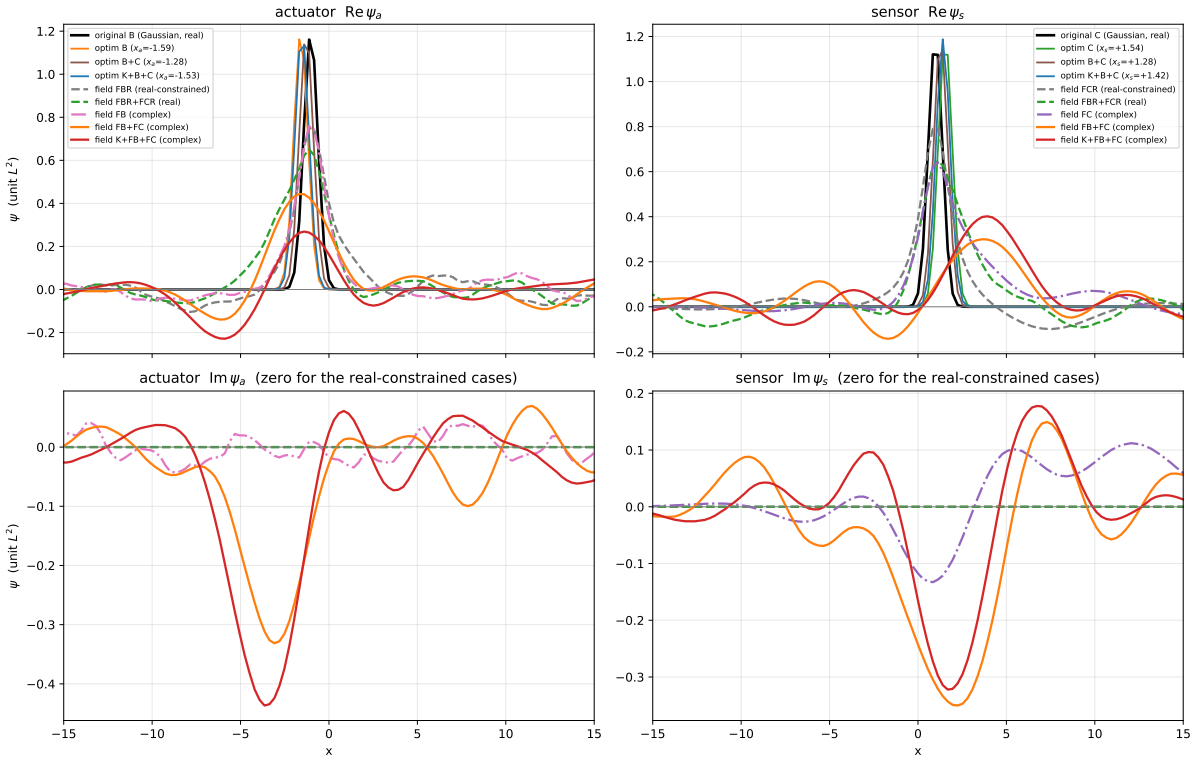



**Figure 2:** Actuator (left) and sensor (right) profiles, all normalised to unit $L^2$ norm so that only *shape* is compared. **Top:** real parts; **bottom:** imaginary parts, which vanish identically for the nominal Gaussians and for the real-constrained cases `FBR`/`FCR`. Placement designs remain narrow Gaussians displaced outwards from the $H_2$-optimal positions; the optimised fields are smooth, broad and multi-lobed, extending far upstream (actuator) and downstream (sensor). The imaginary parts of the complex designs are of the same order as the real parts — that extra freedom is genuinely exploited, and §4.8 separates its contribution from that of shaping alone.

The placement designs (Fig. 2) remain narrow Gaussians, displaced from the nominal $H_2$-optimal positions: the actuator moves *upstream* ($x_a = -1.03 \to -1.59$ for `B`, $-1.53$ for `K+B+C`) and the sensor *downstream* ($x_s = 0.98 \to 1.54$ for `C`, 1.42 for `K+B+C`). The optimiser therefore increases the separation between actuator and sensor relative to the $H_2$-optimal placement — consistent with the peak, rather than integrated, response being penalised.

The optimised *fields* are qualitatively different and considerably more informative. They are smooth — a direct consequence of the Sobolev parametrisation of §3.3, without which they were dominated by grid-scale noise — and they are *broad and multi-lobed*, retaining significant amplitude over 5–10 length units. The optimal actuator extends far **upstream** of the nominal Gaussian, and the optimal sensor far **downstream**. This is exactly what the convective physics suggests: the actuator gains authority by injecting perturbations where they will subsequently be amplified, and the sensor gains observability by integrating the response over the region where amplification has already taken place. A single narrow Gaussian, however well placed, cannot do either, which is the mechanism behind the factor-of-two improvement in Table 3.

## 4.7 Compensator and loop transfer functions

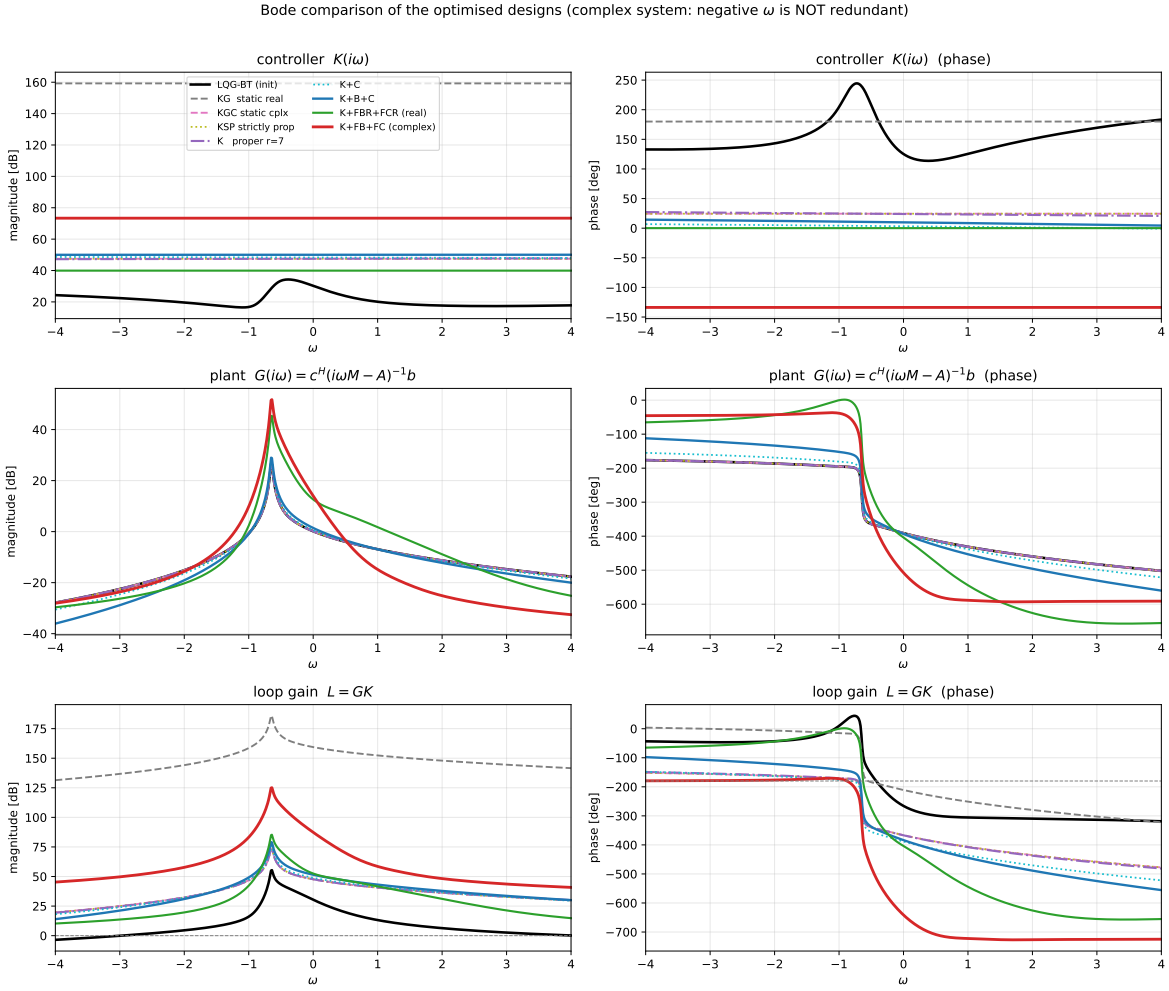



**Figure 3:** Bode comparison. Rows: compensator $K(\mathrm{i}\omega)$; plant $G(\mathrm{i}\omega) = c^H(\mathrm{i}\omega M - A)^{-1}b$ evaluated with each design's own actuator and sensor; loop gain $L = GK$. Because the GL operator is complex, $K(-\mathrm{i}\omega) \neq \overline{K(\mathrm{i}\omega)}$ and the negative-frequency half-plane carries independent information. Note the top-left panel: the LQG-BT reference is the only *resonant* compensator; every optimised design is flat.

The top-left panel of Figure 3 contains the clearest single piece of evidence in this study. Measured over $\omega \in [-4, 4]$:

| design | max $\lvert K\rvert$ [dB] | min $\lvert K\rvert$ [dB] | variation |
|---|---|---|---|
| LQG-BT (reference) | 35.3 | 17.3 | 18.0 dB |
| `KG` (static real) | — | — | 0.00 dB (constant by construction) |
| `KGC` (static complex) | — | — | 0.00 dB (constant by construction) |
| `KSP` (strictly proper, 52 free parameters) | 51.5 | 50.7 | **0.73 dB** |

The LQG-BT compensator is strongly resonant, with 18 dB of variation peaked at the plant resonance: it is trying to invert the plant, which is what minimising the frequency-*integrated* response asks for. Every $H_\infty$-optimised compensator is instead essentially *flat*. The seventh-order strictly-proper design `KSP` was free to use all fifty-two of its parameters and chose a transfer function that varies by less than one decibel across eight units of frequency — it has converged

to a constant without being told to. This is precisely what §4.3 predicts: the only thing the compensator can do is cancel the single rank-one component of the resolvent that is simultaneously reachable by the actuator and visible to the sensor, and a scalar suffices for that.

The plant row shows the sharp GL resonance near $\omega \simeq -0.65$ shared by all designs, displaced and reshaped for the field cases because they use different $b$ and $c$. The loop-gain row shows the consequence of the flat, high-gain compensators: $|L|$ reaches 175 dB at the resonance for the field designs, and the loop phase passes through $-180°$ well away from the crossover, which is consistent with the healthy stability margins of Table 3 ($\alpha$ between $-0.14$ and $-0.50$).

### 4.8 Real versus complex actuator and sensor fields

Splitting the field cases by realness (Figure 2, lower row) separates two effects that were previously conflated:

| design | $\gamma_\infty$ | |
|---|---|---|
| best Gaussian placement, `K+B+C` | 8.4643 | reference |
| `FBR+FCR` (*real* fields) | 5.6097 | $-34\,\%$: shaping alone |
| `FB+FC` (complex fields) | 4.3807 | $-22\,\%$ further: complex freedom |
| `K+FBR+FCR` (real + compensator) | 4.1140 | best fully *real* design |
| `K+FB+FC` (complex + compensator) | 3.6741 | $-11\,\%$ further: complex freedom |

Roughly *two thirds* of the field advantage over placement is available with strictly real profiles, which is the physically realisable comparison against a real Gaussian; the remaining third requires a complex profile, i.e. a distributed actuator and sensor with a prescribed *phase* variation in space. The imaginary parts in Figure 2 are not small, so this is a real degree of freedom being exploited and not a numerical artefact. Which of the two figures is quoted therefore depends on what hardware is envisaged, and the report quotes both.

## 5 Discussion: methodological lessons

Several findings generalise beyond this benchmark and are, in our view, the main contribution of the study.

**Mass consistency is not a detail.** An inconsistent weighting of the disturbance channel does not produce a constant error but destroys mesh convergence. Any weak-form formulation must place $M^{1/2}$ on both sides of the resolvent; the collocation form $M^{-1/2}$ is not merely different, it diverges as $h \to 0$.

**A local $H_\infty$ evaluation inside an optimisation loop is unsafe.** This is the single most important practical lesson. The failure is silent, and it is worse than an inaccurate objective: the optimiser tracks a local maximiser, and as it flattens that peak the true norm can *increase* while the reported value decreases. Certification must be part of the loop, not merely a final audit. The cost is modest — from a good lower bound the certificate converges in two calls — precisely because the cheap local method and the certified global method are complementary.

**Realisation conditioning matters as much as algorithm choice.** The spectrum is realisation-invariant; its conditioning is not. A minimal (companion) parametrisation cost four orders of magnitude in conditioning, broke the certification pencil, and mis-scaled the optimiser. Two additional parameters per state, in tridiagonal form, removed the problem entirely.

**Field optimisation requires regularisation, and the reason is physical.** The resolvent damps high wavenumbers, so those directions are a near-null-space of the objective. Without a Sobolev metric the optimiser deposits grid-scale noise that costs nothing in the objective and that would look entirely different on another mesh. The regularisation changes only the search metric — the objective and the optimum are untouched — and in practice it *improved* the attained norm.

**Diverse initialisation is necessary, and the $H_2$ optimum is a poor start.** LQG-BT minimises the integrated response; $H_\infty$ penalises the peak. From the $H_2$ basin the only local route to a lower peak is often to push a pole towards the imaginary axis, which produces spuriously low norms at vanishing stability margin. $H_\infty$ central compensators built on low-rank exogenous channels are cheap, provide structurally different starting points, and approach the optimum from the interior of the stable region.

# 6 Limitations of the optimisation strategy

The physics results of §4 are certified and, we believe, sound. The *optimiser* is the weakest part of this work and we set out its shortcomings explicitly, because they determine what should be done differently at cavity scale.

## 6.1 What did not converge

Of nineteen designs, nine reach a smooth stationary point (single active frequency, $\|\nabla\gamma\| \sim 10^{-6}$–$10^{-9}$ in the optimiser's own coordinates) and five reach an equalised optimum (two or three active frequencies within 0.03 %). The remaining five — `FB+FC`, `K+B+C`, `K+FB+FC`, `K+FBR+FCR`, `KG+FBR+FCR` — stall with a non-vanishing gradient and a runner-up peak 0.22 % to 1.70 % below the maximum. Their $\gamma$ values are therefore **certified upper bounds**: the certification of §3.6 guarantees each is the true global maximum of $\sigma_1(\omega)$ for that design, but the design itself is not a local optimum. The outstanding margin is bounded by roughly 2 % and none of the conclusions of §4 depends on it.

## 6.2 Why the step-size heuristic is not a methodology

The BFGS-SQP solver requires an initial Hessian $H_0$ scaled to the problem (§3.5); with $H_0 = I$ several cases make *zero* progress. We calibrated $H_0$ from a prescribed relative first step and then ran a fixed ladder of decreasing steps. Two objections stand.

First, the ladder had to be tuned per case. The two failure modes require *opposite* corrections: stalling at a kink (two near-equal peaks) needs a *larger* step so that BFGS accumulates curvature from both branches, while stalling because a step crosses the stability boundary needs a *smaller* one. A fixed schedule cannot distinguish them, and in practice the diagnosis was made by hand from the logs. That is not a reproducible procedure. By contrast a user of a tool such as MATLAB's `systune` performs no such tuning.

Second, the obvious automatic remedies do not work here. PyGRANSO's own `scaleH0` (its default) leaves the hardest case at exactly its starting value, and its `prescaling_threshold` — which is precisely the automatic version of our $H_0$ calibration — fails with a recursion error in the version used. We therefore implemented an adaptive ladder that infers the failure mode from the outcome of each stage (improved → refine; stalled but feasible → enlarge; steps leaving the feasible set → shrink) and stops when the solver's own optimality test passes. It reproduces the hand-tuned result automatically on the cases that previously needed tuning, but it did *not*

rescue any of the five stalled designs. The difficulty is not the schedule; it is that a quasi-Newton line search is being asked to infer a kink structure from function values alone.

### 6.3 A residual defect in warm restarts

The convergence traces of `K+FBR+FCR` and `KG+FBR+FCR` begin at $\gamma = 88.6$ and 48.7 although their stored checkpoints are 4.11 and 4.20. A resume that cannot reproduce its own checkpoint on the first oracle call cannot make progress, which explains why repeated restarts of these two cases have been fruitless. The field warm start was corrected once already (the result files stored the *smoothed* field $\psi = P\xi$ while the optimiser variable is $\xi$, so a warm start applied the Sobolev map twice, perturbing the field by 64 %); the residue affects the *real-constrained* field cases specifically and is not yet understood. It is recorded here as an open defect.

## 7 Conclusions and outlook

We have demonstrated a complete, certified, matrix-free co-design procedure for actuator, sensor and compensator on the linearised Ginzburg–Landau equation. The worst-case energy amplification falls from 858 to 8.46 by optimising actuator and sensor *positions* with a seventh-order compensator, and to 3.67 — a factor of 234 — when they are optimised as spatial *fields*. Three results are worth restating.

**Performance is set by the hardware, not the control law.** The sensor-limited and actuator-limited bounds of §4.3 are attained to five significant figures by eight different designs, and a static *complex* gain matches a fifty-four-parameter seventh-order compensator to thirteen digits. The dynamic compensator `KSP`, free to use fifty-two parameters, converges to a transfer function flat to 0.73 dB.

**Certification is not optional.** A locally convergent evaluation of the $H_\infty$ norm inside the loop fails silently and can make the true norm *increase* while the reported value decreases.

**The bounds should be computed first.** They cost seconds, are independent of any optimisation, and say in advance what a hardware layout can deliver.

### 7.1 Ways forward for the optimiser

Two routes are available, and §6 argues that continuing to engineer step-size schedules is not one of them.

**(a) Epigraph reformulation with an exchange loop.** Replace $\min_\theta \max_\omega \sigma_1$ by

$$\min_{\theta,\,t} \; t \quad \text{s.t.} \quad \sigma_1(\theta, \omega_i) \le t \;\; (\omega_i \in \Omega_k), \qquad \alpha(\theta) + \varepsilon \le 0, \tag{18}$$

with $\Omega_k$ an active-frequency set enlarged by the certification whenever it finds $\sigma_1 > t$. The objective becomes *linear*; all the non-smoothness moves into constraints that are individually *smooth*, and the SQP machinery handles the active set explicitly with multipliers rather than having to discover it from function values. This is a semi-infinite programming formulation and is structurally what a bundle method does.

A specific piece of evidence from this study supports it: the singular-value gap $(\sigma_1 - \sigma_2)/\sigma_1$ measured at every optimum lies between $1.8 \times 10^{-2}$ and $9 \times 10^{-1}$ — nowhere near degenerate. Each $\sigma_1(\theta, \omega_i)$ is therefore genuinely smooth in $\theta$, so the epigraph constraints are well behaved and the analytic gradients already implemented apply unchanged.

**(b) A non-smooth bundle / trust-region method.** This is what `systune` uses (Apkarian & Noll). Subgradients from all active frequencies are accumulated in a bundle and the trust radius is adapted inside the solver, so no external step-size logic exists to tune. It is the reference method for structured $H_\infty$ synthesis and would handle the multi-peak cases (`FB+FC` has six local maxima) far better than gradient sampling driven by a scalar step heuristic.

**Recommendation.** We would try (a) first. It is a much smaller change — the oracles, gradients, certification and bounds are all reused without modification, and only the driver changes — and the measured singular-value gaps say the resulting constraints are smooth. Route (b) is the more powerful and better-established method, and is the fallback if the exchange loop proves slow to add frequencies or if the active set cycles. Neither route touches the part of this work that was expensive to get right: the matrix-free oracles supply $\gamma$, the subgradients at the active frequencies, $\alpha$ and its gradient, mesh-independently, and *every* candidate solver needs exactly that interface. The optimiser is replaceable; the oracles are not.

Both should be validated on Ginzburg–Landau before the cavity, against the certified optima obtained here: the nine converged designs give exact targets, the five stalled ones give a direct measure of whether a new method fixes what the present one cannot, and the bounds of §4.3 provide an independent check that any claimed improvement is physically attainable.

### 7.2 Porting to the cavity

The port itself is mechanical: the finite-element assembly is replaced by a DOLFINX assembly returning sparse $A$, $M$, $b$, $c$; the SciPy sparse LU by a PETSc KSP with MUMPS, reusing the factors for $S$ and $S^H$; and the ARPACK eigensolves by SLEPc Krylov–Schur on the same matrix-free operators. The augmented closed-loop form accommodates the singular $E$ arising from the pressure block, and the extension to multiple actuators and sensors requires only widening $U$ and $V$. Two simplifications are expected: the cavity operator is real, so the certification may exploit spectral symmetry and scan only $\omega \geq 0$; and the tridiagonal compensator parametrisation, chosen here partly with that case in mind, handles the migration of real poles into complex-conjugate pairs without any change of parametrisation.

## References


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